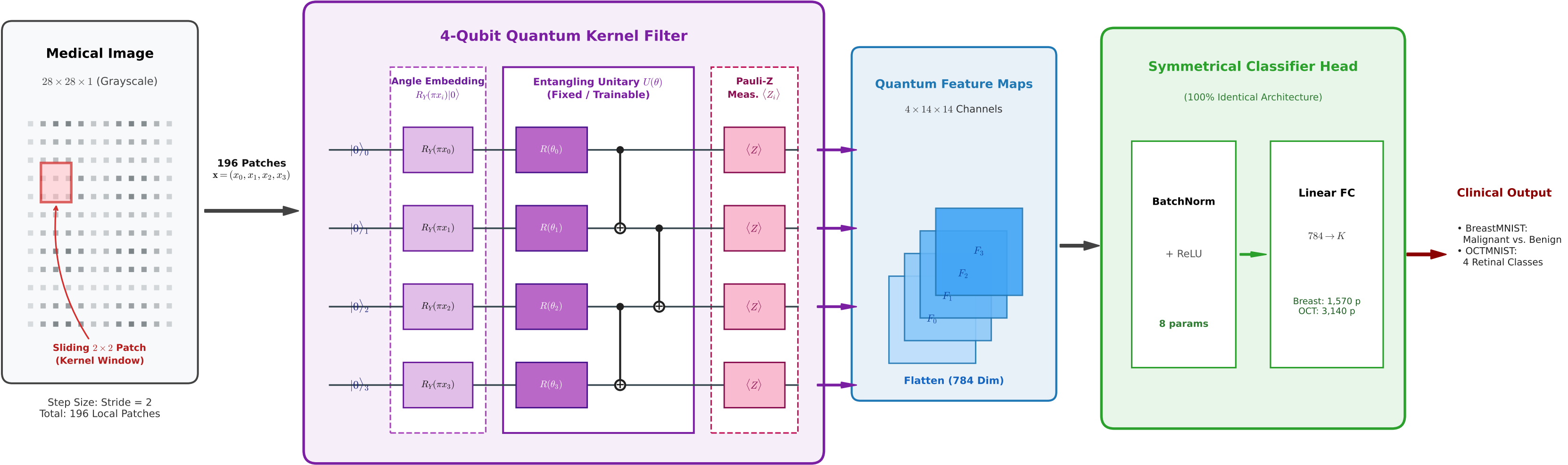


Figure 1: End-to-End Architecture of the Quanvolutional Neural Network (Quanvolution)

Symmetrical 4-Qubit Quantum Kernel vs. Minimum Classical Convolutional Filter Baseline



Classical Baseline (Symmetrical Minimum CNN):

$\text{Conv2D}(c_{\text{in}} = 1, c_{\text{out}} = 4, k = 2, s = 2)$ (20 params)

$\rightarrow 4 \times 14 \times 14$ Feature Maps

$\rightarrow$ Identical Classifier Head

★ Key Fairness Guarantee: Same parameter count in head, same feature dimension (784), isolated feature extractor comparison.